CODEALPHA VIRTUAL INTERNSHIP PROGRAM
ARTIFICIAL INTELLIGENCE PROJECTS

AN INTERNSHIP PROJECT REPORT

Submitted by

K SRI JANANI

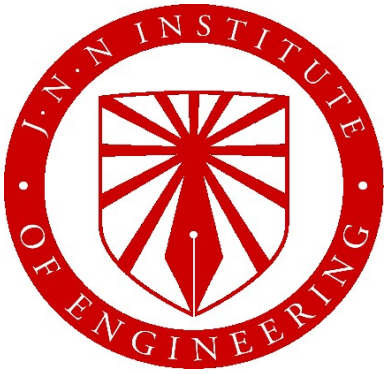
(Reg No: 110725102244)

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

Artificial Intelligence & Data Science



J.N.N Institute of Engineering

(An Autonomous Institution, Affiliated to Anna University, Chennai)

AUGUST - 2026

VISION OF THE INSTITUTE:

Lead the transformation of engineering and management learning experience to educate the next generation of innovators and entrepreneurs who want to make the world a better place.

MISSION OF THE INSTITUTE:

IM1: To develop the required resources and infrastructure and to establish a conducive ambience for the teaching-learning process.

IM2: To nurture professional and ethical values in the students and to instils in them a spirit of innovation and entrepreneurship.

IM3: To encourage a desire for higher learning and research in the students and to equip them to face global challenges.

IM4: To provide opportunities for students to learn job-relevant skills to make them industry ready.

IM5: To interact with industries and other organization to facilitate transfer of knowledge and know-how.

VISION OF THE DEPARTMENT:

To impart quality-education, inculcate professionalism and enhance the problem-solving skills of the students in the domain of Artificial Intelligence & Data Science by applying recent technological tools and incorporating collaborative principles with a focus to make them industry ready.

MISSION OF THE DEPARTMENT:

DM1: To enhance the knowledge of the students with most recent advancements and refresh their insights in the field of Artificial Intelligence and Data Science.

DM2: To equip the students with strong fundamental concepts, analytical capability, programming and problem-solving skills.

DM3: To make the students industry ready and to enhance their employability through training, internships and real-time projects.

DM4: To guide the students to perform research on Artificial Intelligence and Data Science, with the aim to provide solutions to the problems of the industry.

PROGRAM OUTCOMES

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO1	Engineering knowledge	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/Development of solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern Tool usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The Engineer and society	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice
PO7	Environment and sustainability	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES (PEO)

S.No	PROGRAM EDUCATIONAL OBJECTIVES (PEO)
PEO1	To provide graduates with the proficiency to utilize the fundamental knowledge of basic sciences, mathematics, Artificial Intelligence, data science and statistics to build systems that require management and analysis of large volume of data.
PEO2	To enrich graduates with necessary technical skills to pursue pioneering research in the field of AI and Data Science and create disruptive and sustainable solutions for the welfare of ecosystems.
PEO3	To enable graduates to think logically, pursue lifelong learning and collaborate with an ethical attitude in a multidisciplinary team

PROGRAM SPECIFIC OBJECTIVES (PSOs)

S.No	PROGRAM SPECIFIC OBJECTIVES (PSOs)
PSO1	Graduates should be able to evolve AI based efficient domain specific processes for effective decision making in several domains such as business and governance domains.
PSO2	Graduates should be able to arrive at actionable Fore sight, Insight , hind sight from data for solving business and engineering problems
PSO3	Graduates should be able to create, select and apply the theoretical knowledge of AI and Data Analytics along with practical industrial tools and techniques to manage and solve wicked societal problems.

BONAFIDE CERTIFICATE

Certified that this internship project report "CODEALPHA VIRTUAL INTERNSHIP PROGRAM - ARTIFICIAL INTELLIGENCE PROJECTS" is the bonafide work of K SRI JANANI (Reg No: 110725102244), who carried out the internship at CodeAlpha from 20 July 2026 to 20 August 2026 under the supervision of Ms. M. Gayathridevi.

Signature of the HOD with date

Dr. P. NAGARAJAN
Professor and Head,
Department of Artificial Intelligence & Data Science,
J.N.N Institute of Engineering, Chennai.

Signature of the Supervisor with date

Ms. M. Gayathridevi
Project Supervisor,
Department of Artificial Intelligence & Data Science,
J.N.N Institute of Engineering, Chennai.

Certified that the above student has attended the viva voce examination held on .....

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

First and foremost, I express my sincere gratitude to our beloved **FOUNDER & CHAIRMAN Shri. S. JAYACHANDRAN B.Sc., B.L.,** and heartfelt thanks to our **VICE CHAIRMAN Mr. NAVEEN JAYACHANDRAN M.S.,** for providing the necessary facilities and an encouraging academic platform for the completion of my internship projects.

I thank our **PRINCIPAL Dr. G. MOHANBABU, Ph.D.,** for his encouragement and for providing a congenial academic environment that supports practical learning and the development of technical skills.

I take this opportunity to express my profound gratitude to our **HEAD OF THE DEPARTMENT Dr. P. NAGARAJAN** and all the faculty members of the Department of ARTIFICIAL INTELLIGENCE AND DATA SCIENCE for their continuous encouragement, academic support and valuable advice during the internship period.

I extend my sincere thanks to my college supervisor **Ms. M. Gayathridevi,** Department of Artificial Intelligence and Data Science, J.N.N Institute of Engineering, for her guidance, constructive suggestions and support in preparing this internship report.

I sincerely thank **CodeAlpha** for providing the opportunity to undertake the Artificial Intelligence Virtual Internship from 20 July 2026 to 20 August 2026, under internship ID CA/DF1/213728. The Language Translation Tool and FAQ Chatbot enabled me to apply web development concepts to practical language-assistance and information-access interfaces. I also appreciate the CodeAlpha team for the internship opportunity and the official documents recording its completion.

Finally, I would like to thank my parents and friends for their constant support, motivation and encouragement throughout the internship and the preparation of this report.

K Sri Janani
110725102244

ABSTRACT

Artificial intelligence applications and modern web development offer practical ways to improve access to language assistance and frequently requested information. This report documents the objectives, methodology, interface implementation and observed results of the Artificial Intelligence Virtual Internship undertaken by K Sri Janani at CodeAlpha from 20 July 2026 to 20 August 2026, under internship ID CA/DF1/213728. The primary objective was to apply front-end development concepts to clear, interactive browser-based applications.

The internship consisted of two practical tasks. Task 1 involved developing a Language Translation Tool with source and target language selectors, a text-entry area, a Translate button and a separate result section containing a Copy control. The project README identifies the MyMemory Translation API as the translation service. The supplied execution screenshot demonstrates English input "hi" translated into a Tamil greeting, showing a specific input-to-output interaction.

Task 2 involved developing an FAQ Chatbot with a conversation area, a question-input field, a Send button and quick-question controls. The interface distinguishes user questions from assistant answers through message alignment and contrasting colors. The supplied output demonstrates responses to "What is CodeAlpha?" and "How can I contact support?", illustrating how a conversational interface can present commonly requested information.

Both projects use HTML, CSS and JavaScript, with Visual Studio Code and GitHub supporting development and project organization. Repository screenshots show the separate task folders and their index.html, style.css, script.js and README.md files. The report explains the project requirements, interaction workflows, interface components and learning outcomes, supported by execution screenshots, repository evidence and official internship documents.

Together, these two projects connect academic learning with practical work in interface design, programming, debugging and technical documentation. The observed outputs demonstrate selected application behavior, while broader translation quality, FAQ coverage, error handling and accessibility remain areas for further evaluation. Future development should prioritize reproducible tests, clear processing and failure feedback, improved accessibility and continued maintenance of the project documentation.

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Evidence Included in the Report

The report contains execution screenshots for both internship tasks and GitHub repository screenshots showing their project files. The Language Translation Tool README identifies the MyMemory Translation API. Official internship documents are reproduced in the appendices.

The academic report uses K Sri Janani. The official documents use Sri Janani K and are reproduced with their original wording and appearance.

CHAPTER 1
INTRODUCTION

1.1 Overview

Web applications allow users to interact with software through a browser. A successful interface turns a general task into a clear sequence of actions: entering information, selecting an option, requesting an operation and understanding the result. Language assistance and frequently asked questions are useful project contexts because they place these interaction requirements at the center of development.

During the CodeAlpha Virtual Internship Program, I completed two web-based projects: a Language Translation Tool and an FAQ Chatbot. The first organizes text input and language selection around a translation workflow. The second organizes questions and answers as a conversation, with quick-question controls for common topics. Although the internship domain is Artificial Intelligence, the supplied material primarily documents the front-end interfaces and a chatbot response.

1.2 Internship Details

Field	Details
Student	K Sri Janani
Register number	110725102244
Degree and branch	B.Tech Artificial Intelligence & Data Science
Institution	J.N.N Institute of Engineering
Organization	CodeAlpha
Internship domain	Artificial Intelligence
Student ID	CA/DF1/213728
Period	20 July 2026 to 20 August 2026
Mode	Virtual internship
College supervisor	Ms. M. Gayathridevi
Head of Department	Dr. P. NAGARAJAN

INTRODUCTION CONTINUED

1.3 Objectives

• Apply HTML, CSS and JavaScript to interactive browser interfaces.

• Organize text input, language selection and result display in a translation tool.

• Present frequently asked questions through a readable conversation interface.

• Use quick-question controls to reduce the effort required to submit common queries.

• Develop practical familiarity with debugging, project organization and GitHub.

• Document project outcomes clearly and distinguish observed behavior from proposed improvements.

1.4 Overview of the Two Tasks

Task 1, Language Translation Tool, provides From and To language controls, text input, a Translate button, a result section and a Copy button. The supplied execution screenshot shows English input "hi" and a Tamil greeting in the result area.

Task 2, FAQ Chatbot, provides assistant and user messages, an input field, a Send button and Quick Questions. The screenshot demonstrates responses about CodeAlpha and contacting its support team.

1.5 Scope of the Report

The report covers background concepts, methodology, interface implementation, results and future work for two tasks. It uses project execution screenshots, repository screenshots and official internship documents. The translation README identifies MyMemory as the translation API. Screenshots document selected results; source-code inspection and wider testing would be needed to assess internal logic and reliability.

INTRODUCTION - EXTENDED DISCUSSION

1.6 Problem Definition

Language differences can make it difficult for users to understand short messages or reuse information in another language. A translation tool addresses this need through a clear text-entry and language-selection workflow. A second problem is repeated information requests: users often need the same basic details about an organization or its services. An FAQ chatbot presents these answers in a conversation instead of requiring the user to search through a long document. These two problems formed the practical context of the internship tasks.

1.7 Intended Users and Use Cases

The Language Translation Tool is suitable as an educational prototype for learners and users who need assistance with short text. Its visible English and Tamil selections demonstrate a language pair relevant to multilingual communication. The FAQ Chatbot supports visitors looking for introductory information about CodeAlpha and guidance on contacting support. In the supplied output, both questions receive a readable answer. These use cases explain the interface requirements without implying a production deployment or unrestricted coverage.

1.8 Functional Requirements

Task 1 requires a source-language control, a target-language control, a text area, a translation action and a separate output region. The result region includes a Copy control so that the translated text can be reused. Task 2 requires a question field, a Send action, a conversation area and quick-question options. User messages and assistant answers should remain distinguishable. The repository images show separate folders and HTML, CSS, JavaScript and README files for each task, supporting an organized project structure.

1.9 Quality Requirements

A useful application should present clear labels, readable text and a predictable sequence of actions. It should retain enough context for users to understand the result of their request. Reliability involves handling unavailable services and invalid input without leaving a misleading result on screen. Accessibility involves keyboard navigation and understandable feedback. These quality requirements provide a basis for future review; the screenshots establish selected interface states and outputs rather than completion of every requirement.

CHAPTER 2
LITERATURE REVIEW

2.1 Web Development and Front End Technologies

HTML defines the structure of a web page, including headings, form controls and content areas. CSS controls presentation, such as spacing, typography, colors and layout. JavaScript supports interactions by responding to events and updating the displayed page. Separating these responsibilities makes a small application easier to understand and maintain.

Both internship projects use form-like input followed by an action and a response. In the translation tool, the response is presented in a dedicated result area. In the chatbot, new questions and answers are presented in the conversation area. This difference changes the interface structure even when both projects use the same front-end technologies.

2.2 Language Translation Interfaces

A translation interface captures text and a language pair. The source selection identifies the language of the input, and the target selection identifies the requested output language. A result area separates translated text from the original entry, while a Copy control helps users reuse the result.

A translation interface can connect to an external service or a locally implemented system. The supplied Language Translation Tool README identifies the MyMemory Translation API. The execution screenshot shows one English-to-Tamil greeting translation. Evaluation across additional inputs should consider meaning, names, punctuation, context and the selected language pair.

2.3 FAQ Chatbots

An FAQ chatbot presents answers to common information requests through a conversational interface. Some systems use fixed question-and-answer entries; others apply more advanced retrieval or language processing. The visible screenshot alone cannot identify the matching method. For this report, the demonstrated behavior is the presentation of a CodeAlpha question and answer, supported by quick-question choices.

LITERATURE REVIEW CONTINUED

2.4 Interaction Design and Usability

Labels, action names and feedback help users understand an application. The translation tool uses From and To labels and a clear Translate action. The chatbot distinguishes user and assistant messages using alignment and contrasting bubble colors. Quick questions expose available topics without requiring users to guess suitable wording.

Usability also depends on states that are not fully represented in the supplied screenshots. These include empty input, processing, unavailable results and failures. A completed interface should explain these states using readable text, preserve user input where appropriate and allow recovery without reloading the page. These are design considerations and proposed improvements rather than claims of verified implementation.

2.5 Tools and Their Purpose

Tool or technology	Role described in the supplied report
HTML	Page structure, labels, controls and content sections.
CSS	Layout, color, typography and visual presentation.
JavaScript	User interaction and page updates.
Visual Studio Code	Editing and organizing project files.
GitHub	Source-code management and project upload.
Web browser	Displaying the interfaces and checking their behavior.

2.6 Reliability and Accessibility

Reliable interaction requires input checks and useful feedback. Accessible interaction requires readable contrast, descriptive labels and keyboard operation. For a translation result or incoming chatbot answer, an accessible status announcement can help screen-reader users understand that the page has changed. These improvements should be checked through targeted tests rather than assumed from visual appearance.

For both projects, documentation should record what was tested, the observed output and any unresolved limitations. A screenshot is useful evidence of a particular state, but it does not establish behavior across every input, browser or device.

LITERATURE REVIEW - EXTENDED DISCUSSION

2.7 Translation Services and API-Based Applications

A translation API separates the browser interface from the service that produces translated text. The browser collects the input and language pair, while the service processes the request and returns a result for presentation. The supplied project README names the MyMemory Translation API. This establishes the documented service choice, although it does not expose the actual request code or configuration. An API-based application must consider the quality of both the remote result and the local interface that displays it.

2.8 Translation Quality and Context

Translation quality depends on meaning as well as on whether the interface displays text. A short greeting can demonstrate a working interaction, but longer sentences may introduce ambiguity, idioms, grammatical differences or names that require special handling. English and Tamil differ in sentence structure and writing system, making contextual review useful. The observed greeting is therefore a concrete demonstration of output, while evaluation across a larger collection of examples remains a separate activity. A reviewer should compare the intended meaning with the displayed translation.

2.9 Conversational Information Access

An FAQ interface organizes information around a user question and a corresponding response. This format reduces the need to scan unrelated content and allows common topics to be offered as shortcuts. The supplied chatbot shows questions about CodeAlpha and support, with assistant answers beneath them. A conversational layout also needs sensible limits: if a question is outside the available information, a helpful fallback should guide the user to supported topics. The exact matching strategy cannot be determined from the visible interface alone.

2.10 Separation of Interface and Application Logic

HTML, CSS and JavaScript have distinct responsibilities in a browser application. HTML identifies controls and content regions, CSS makes the layout readable, and JavaScript connects actions to updates. Separate project files make those responsibilities easier to inspect and maintain. A visual change should not unintentionally alter how a question is processed, and a logic change should preserve understandable output. The repository file lists support this separation as a documented structure, while further code review would be needed to assess its implementation in detail.

CHAPTER 3
METHODOLOGY

3.1 Overall Approach

The original report describes a workflow of understanding requirements, designing interfaces, coding, testing, debugging and checking outputs. Each task was organized around a small set of visible user actions. This kept the development focus on a clear input-to-output interaction rather than an unnecessarily complex interface.

3.2 Language Translation Tool Design

The translation layout groups the language controls above the text-entry area. The Translate button follows the input, and the result appears in a separate section underneath it. This arrangement supports a reading order of selection, entry, action and result. The Copy control belongs with the result because it acts on output rather than on source text.

3.3 FAQ Chatbot Design

The chatbot separates the conversation history from the question-entry controls. A greeting invites a question, user and assistant bubbles distinguish speakers, and quick questions provide direct routes to common topics. Direct entry and quick-question selection should lead to a consistent response presentation.

3.4 Development and Review

According to the original report, HTML and CSS were used to construct and style the interfaces, while JavaScript managed the interactions. Visual Studio Code supported editing and GitHub supported project upload. Debugging and repeated testing are reported as part of the work, but individual test logs were not supplied.

3.5 Evidence and Evaluation Method

The report evaluates the supplied execution and repository screenshots. The translation output shows English input "hi" and a Tamil greeting; the FAQ output shows answers about CodeAlpha and contacting support. The translation README identifies MyMemory as the API. Chapter 5 separates these observed results from proposed wider validation. No measured accuracy or response-time benchmark is reported.

METHODOLOGY - EXTENDED DISCUSSION

3.6 Requirement-to-Interface Mapping

A structured development approach begins by translating each requirement into a visible control or response. For the translation task, language choice maps to two selectors, source content maps to a text area, and the requested operation maps to the Translate button. For the chatbot, a request maps to a question field or quick-question selection, and its outcome maps to a new assistant message. This mapping helps reviewers understand the purpose of each component and reduces the chance of adding controls without a clear role.

3.7 Translation Interaction Workflow

The visible workflow begins with selecting English as the source and Tamil as the target, entering "hi", and requesting translation. The result screenshot records a Tamil greeting in the output area. The README documents API-based translation using MyMemory. Conceptually, the application collects the text and language pair, submits a request, receives a response and updates the result region. Input validation, a processing indicator and recovery from service errors are recommended review points. Their presence is not established by the single execution screenshot.

3.8 FAQ Interaction Workflow

The chatbot presents a greeting, allows the user to ask a question and places the answer in the conversation area. Quick Questions provide an alternative entry route for common topics. The demonstrated session includes an organizational question followed by a support question, each with an associated answer. A consistent workflow should use the same answer presentation for typed and selected questions. The visible scrollbar accommodates the growing conversation, while the input field and Send control remain in a separate region underneath it.

3.9 Documentation and Evidence Collection

The screenshots serve different documentation purposes. Execution screenshots capture the interface together with selected inputs and outputs; repository screenshots capture project folders, file names and README information. Official documents establish internship details and completion. These sources should be interpreted together, with each supporting the claims it actually records. Future evidence collection should note the input, language pair or question, expected behavior, observed result and any error. Recording those conditions would make later verification easier to repeat and compare.

CHAPTER 4
IMPLEMENTATION

4.1 Task 1 Language Translation Tool

4.1.1 Objective and Interface Components

The objective was to provide a browser interface through which a user can enter text, choose languages and request a translation. The supplied output image shows a central panel with a descriptive title, a short instruction line and a clear sequence of controls. English is selected in the From field and Tamil in the To field.

Component	Purpose
From selector	Identifies the source language.
To selector	Identifies the requested output language.
Text area	Accepts the text to be translated.
Translate button	Initiates the requested operation.
Translation Result section	Separates the output from the input.
Copy button	Provides an action for reusing output text.

4.1.2 Interface Organization

The two language selectors appear side by side, so their relationship is visible before the user enters text. The text area provides room for more than a single line. The wide Translate button makes the primary action easy to locate. The result section uses a separate background to distinguish output from editable input.

4.1.3 Implementation Evidence

The supplied repository screenshot lists index.html, style.css, script.js and README.md for the Language Translation Tool. Its README identifies HTML, CSS, JavaScript and the MyMemory Translation API. Figure 4.1 documents a displayed translation result; Figure 4.3 records the project structure and documented features. The API integration code is not visible in these screenshots.

LANGUAGE TRANSLATION TOOL OUTPUT

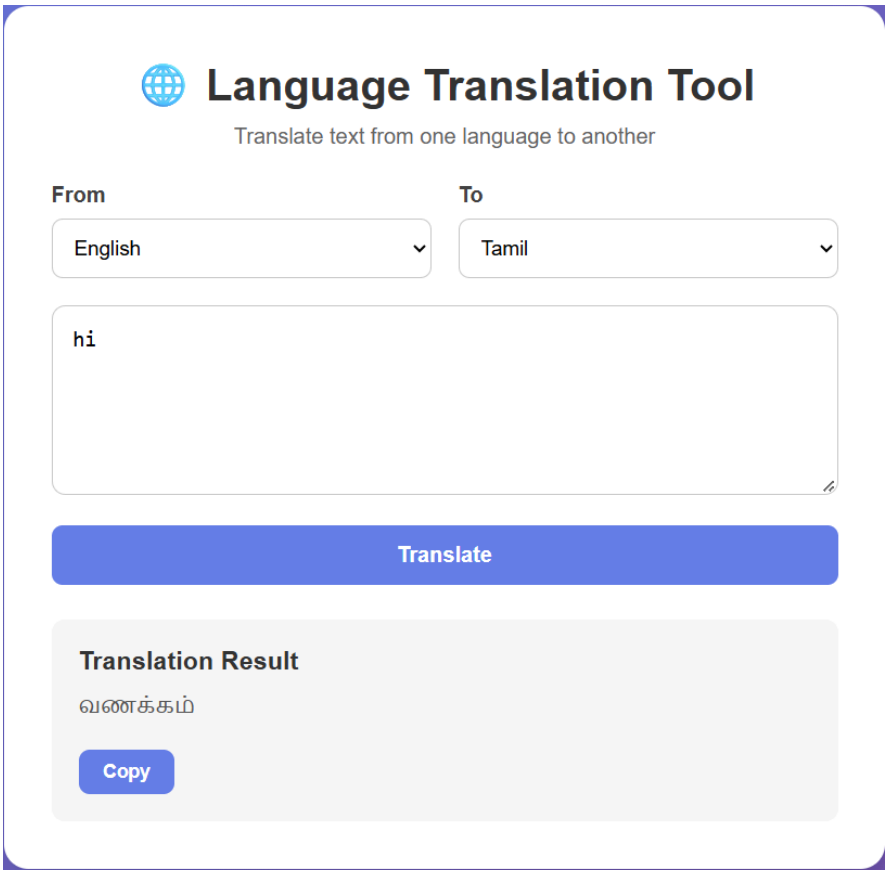


Fig. 4.1 - English-to-Tamil translation execution result

The input "hi" is translated from English into a Tamil greeting. The result is displayed below the Translate button, beside the Copy control.

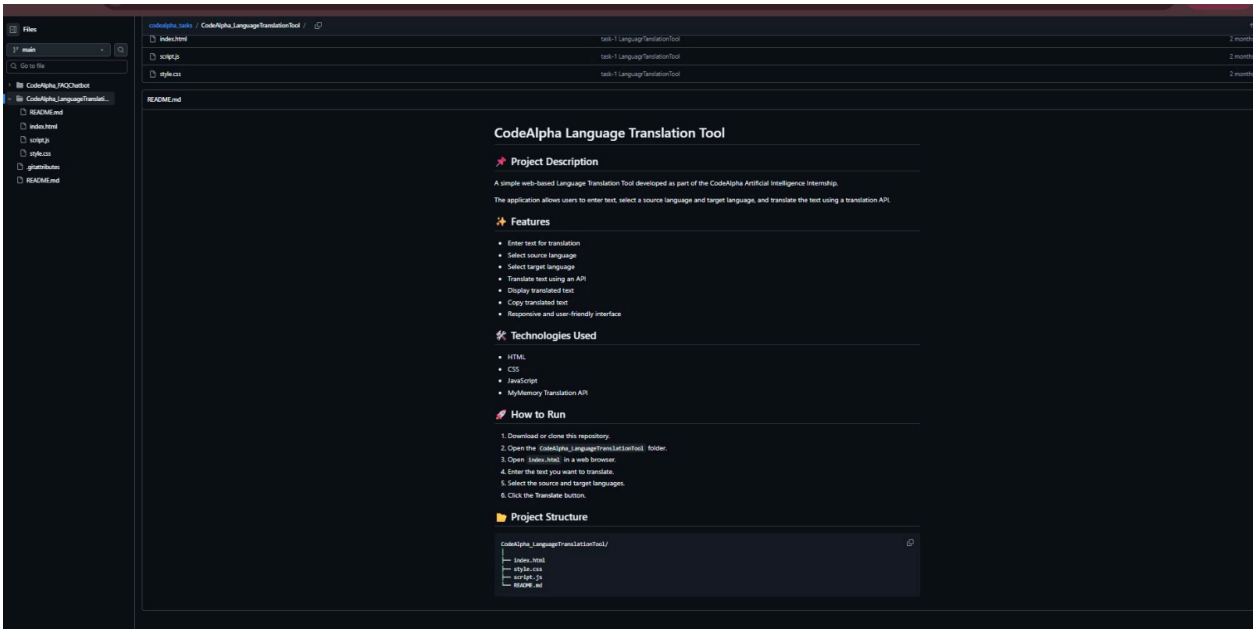


Fig. 4.3 - Language Translation Tool repository and README

FAQ CHATBOT IMPLEMENTATION

4.2 Task 2 FAQ Chatbot

4.2.1 Objective and Interface Components

The objective was to provide answers to frequently asked questions through a simple chat interface. The supplied screenshot shows a greeting, a user question and an assistant answer. A text-entry field and Send button support direct questions, while a Quick Questions section provides four visible topic choices.

Component	Purpose
Greeting	Introduces the assistant and invites a question.
Conversation area	Displays user questions and assistant answers.
Question input	Accepts a directly typed question.
Send button	Submits the entered question.
Quick Questions	Offers common topics as selectable prompts.

4.2.2 Message Presentation

The user question appears in a blue bubble aligned toward the right, while the assistant greeting and answer appear in gray bubbles. This visual distinction helps readers follow the exchange. The answer is displayed beneath the corresponding question, and the input controls remain separate from the conversation content.

4.2.3 Question Topics

The visible quick questions are “What is CodeAlpha?”, “How can I contact support?”, “What tasks are available?” and “What technologies can I use?” These choices reduce typing and communicate the intended information scope. Only the first topic’s response is visible in the screenshot; responses to the other topics are not documented by that image.

4.2.4 Implementation Evidence

The repository screenshot shows the CodeAlpha_FAQChatbot folder containing README.md, index.html, script.js and style.css. The execution screenshot documents two questions with corresponding answers. Figure 4.2 shows this conversation and Figure 4.4 shows the project files. The internal question-matching logic is not visible in the supplied repository image.

FAQ CHATBOT OUTPUT

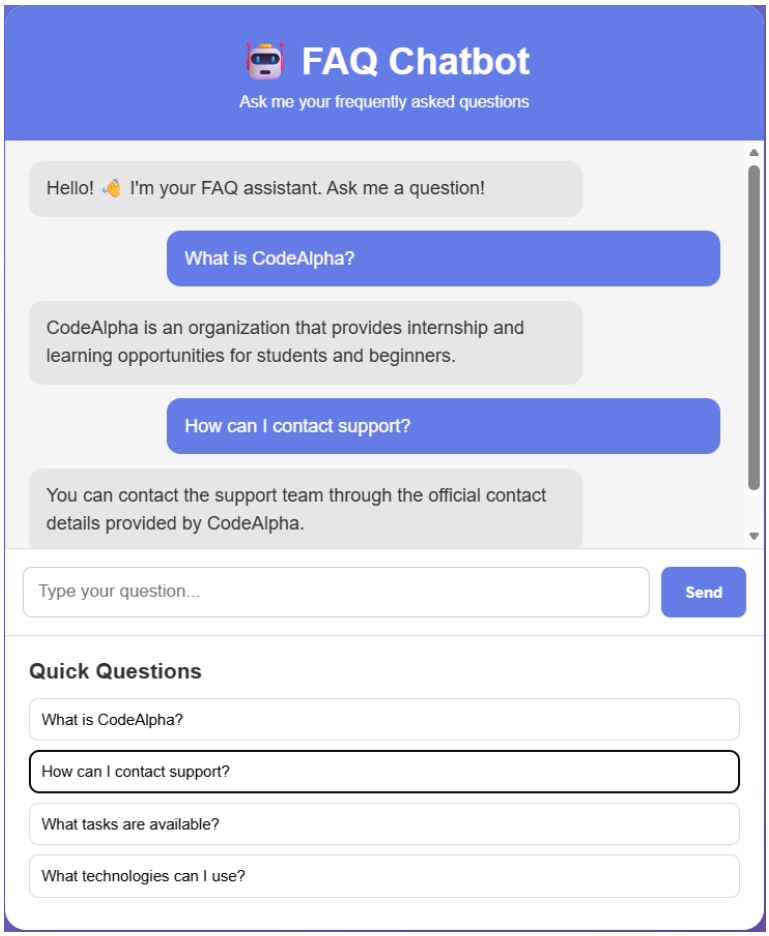


Fig. 4.2 - FAQ Chatbot execution showing two questions and answers

The chatbot responds to "What is CodeAlpha?" and "How can I contact support?". Both questions and their corresponding answers are visible in the conversation.

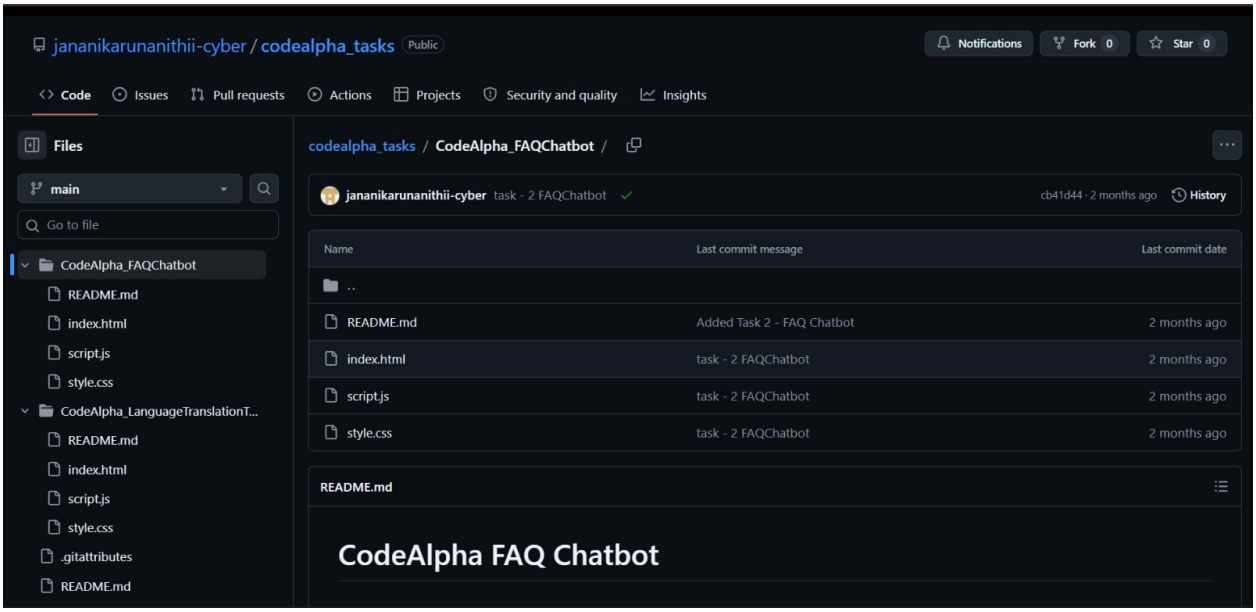


Fig. 4.4 - FAQ Chatbot repository and project files

IMPLEMENTATION - EXTENDED DISCUSSION

4.3 Project Organization and File Responsibilities

Both task folders contain index.html, style.css, script.js and README.md in the supplied repository images. The HTML file serves as the browser entry point, the stylesheet supports presentation, the script file holds interaction code, and the README explains the project. This arrangement gives each task its own development boundary and makes it easier to locate files during review. The screenshots document the names and structure of these files; they do not provide a complete source listing or demonstrate every internal function.

4.4 Language Translation Tool Components

The translation output demonstrates a vertically arranged interaction. The two language controls appear first, followed by text input, the Translate action and the result panel. Separating source text from output reduces confusion about which content was entered and which content was produced. The Copy button is placed inside the result panel, making its intended target apparent. The README lists entering text, selecting languages, translating through an API, displaying translated text and copying output as project features. Only the displayed translation is directly demonstrated in the execution image.

4.5 FAQ Chatbot Components

The FAQ output uses alignment and color to separate the speakers. User questions appear in blue bubbles aligned toward the right, while assistant text appears in lighter bubbles toward the left. The session records two topics, so the image demonstrates that the conversation can contain more than one question-and-answer pair. The input and Send control form a distinct interaction row. Quick-question buttons expose four topics: CodeAlpha, support, available tasks and technologies. The image shows answers for the first two topics, while the other responses remain outside the supplied evidence.

4.6 Error Handling and Maintainability

The next implementation review should examine the boundary between user actions and application state. Empty translation input should produce guidance; a failed service request should not leave stale output that looks current. An empty chatbot message should not create a blank conversation entry, and an unsupported question should receive clear guidance. Keeping these checks in understandable functions would make changes easier to review. These are proposed improvements and review criteria, rather than additional features claimed from the screenshots. The same approach applies to labeling, keyboard access and readable status messages.

CHAPTER 5
RESULTS AND DISCUSSION

5.1 Summary of Observed Results

Project	Observed evidence	Unverified behavior
Language Translation Tool	English "hi" translated to a Tamil greeting; language selectors and Copy control.	Broader translation accuracy, copying and failure handling.
FAQ Chatbot	Displayed answers about CodeAlpha and contacting support; input and quick questions.	Matching algorithm, other answers, fallback behavior and broad query coverage.

Both projects demonstrate an input-to-output interaction. The translation screenshot shows English input "hi" and a Tamil greeting as the result. The FAQ screenshot shows answers to two selected questions. The GitHub screenshots provide repository evidence for both tasks. These observations establish selected outputs, while broader accuracy and error handling require additional testing.

5.2 Proposed Translation Validation Cases

Case	Expected behavior to verify
Empty text	Explain that text is required before translation.
English to Tamil sentence	Display output whose meaning is checked by a fluent reviewer.
Punctuation and names	Preserve or appropriately handle punctuation and proper names.
No available result	Show a clear message without presenting stale output as current.
Copy action	Copy the displayed result and confirm the action.

These cases are proposed checks, rather than a complete executed test log. The screenshot demonstrates one translation result. Broader translation evaluation requires representative input-output pairs and review of meaning; one displayed greeting does not establish an overall accuracy percentage.

RESULTS AND DISCUSSION CONTINUED

5.3 Proposed FAQ Validation Cases

Case	Expected behavior to verify
Displayed CodeAlpha question	Present the intended answer consistently.
Other quick questions	Return the appropriate answer for each visible topic.
Unknown question	Provide a clear fallback or guidance to supported topics.
Empty input	Avoid adding an empty message to the conversation.
Long question or answer	Wrap content without obscuring input controls.
Keyboard interaction	Allow controls to be reached and operated through the keyboard.

5.4 Skills and Challenges

The original report identifies practical learning in web development, coding, debugging, GitHub usage and project development. It also reports coding errors, implementation difficulties and GitHub upload problems, addressed through debugging, checking errors, consulting resources and repeated testing. Detailed incident logs were not supplied, so specific error messages and fixes are not invented here.

The two projects reinforce the value of clear control labels, consistent visual hierarchy and direct feedback. They also illustrate a documentation challenge: visible interface completion must be distinguished from the correctness of underlying behavior. Recording input, observed response and test conditions would make later project assessment more reproducible.

5.5 Limitations and Improvement Priorities

The evidence comprises the written report, two execution screenshots, two GitHub screenshots and official documents. The translation example and README document a result and identify the API. Further checks should cover longer translations, all FAQ topics, unknown questions, error recovery and accessibility. Source-code review and reproducible tests would strengthen the evaluation.

RESULTS AND DISCUSSION - EXTENDED ANALYSIS

5.6 Interpretation of the Translation Result

The translation screenshot records an observable input-output example: the source is English, the target is Tamil, and the entered text is "hi". A Tamil greeting appears in the result section. This is stronger evidence than an empty interface because the output region contains a response to a specific input. The result supports the conclusion that a selected translation interaction was demonstrated. It does not, by itself, establish sentence-level accuracy, support for every language option, response-time performance or correct behavior when the service is unavailable.

5.7 Interpretation of the Chatbot Results

The FAQ screenshot records two exchanges. The first answer describes CodeAlpha as an organization providing internship and learning opportunities. The second answer directs the user to official contact details for support. These exchanges show the relationship between questions and responses and demonstrate presentation of a conversation history. The available evidence does not reveal how the question was matched to its answer or whether differently worded questions receive the same response. The remaining quick questions should be checked individually before broader coverage is claimed.

5.8 Comparison of the Two Tasks

The tasks share a browser-based interface and an input-action-result pattern, but their outputs have different meanings. Translation aims to preserve the meaning of source text in another language, whereas an FAQ answer provides relevant information about a supported topic. Translation review therefore focuses on language and context, while FAQ review focuses on relevance and consistency. Both require readable controls and appropriate feedback. Developing the two interfaces together illustrates how a common set of web technologies can support different interaction requirements.

5.9 Learning Outcomes and Further Evaluation

The project work supports practical understanding of page structure, styling, event-driven interaction, file organization and technical reporting. The repository images connect the visible interfaces with documented project folders, while the execution images provide examples that can be explained in the report. A useful next evaluation would record several translation inputs and all four FAQ topics, followed by empty-input, unsupported-input and recovery cases. Results should be reported individually before any overall metric is introduced. This would strengthen the relationship between implementation, evidence and the conclusions drawn from the work.

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 Conclusion of Individual Tasks

The Language Translation Tool provided experience with language selection, text input and result presentation. The execution screenshot demonstrates an English-to-Tamil greeting translation, and the repository README identifies the MyMemory Translation API.

The FAQ Chatbot provided experience with a conversation interface and common-question controls. The execution screenshot demonstrates answers to questions about CodeAlpha and support contact information. The repository screenshot records the project files.

6.2 Overall Conclusion

The CodeAlpha Virtual Internship Program in Artificial Intelligence, completed from 20 July 2026 to 20 August 2026, provided practical exposure to web-based project development. The two projects apply HTML, CSS and JavaScript to user-facing interactions and support learning in debugging, organization and documentation. The official certificate and recommendation letter confirm successful completion of the internship.

The projects offer a foundation for further work in language and information-assistance applications. A careful distinction between an interface, its underlying mechanism and evaluated results is essential when extending them. This report records the available evidence while identifying the checks needed to support stronger conclusions.

6.3 Future Work

• Document the translation mechanism and supported languages, then evaluate representative translations with a fluent reviewer.

• Expand FAQ coverage and define how unmatched questions are handled without fabricating information.

• Add processing, success and failure feedback where needed, including confirmation for copying.

• Improve keyboard access, labels, contrast and responsive layouts through targeted accessibility checks.

• Maintain source documentation and reproducible test records alongside project updates.

CONCLUSION - DEVELOPMENT ROADMAP

6.4 Consolidated Internship Contribution

The internship involved two complementary tasks: a Language Translation Tool and an FAQ Chatbot. Together they provide a practical setting for applying HTML, CSS and JavaScript to interactive browser applications. The translation demonstration connects language selection and entered text with a displayed result. The chatbot demonstration connects questions with information presented in a conversation. The GitHub images show how both tasks were organized and documented, and the official internship documents remain part of the report record.

6.5 Priority 1: Functional Verification

The first improvement priority is reproducible verification. Translation cases should include a greeting, a sentence, punctuation and a proper name, with the selected language pair recorded for each case. A fluent reviewer should assess whether meaning is retained. FAQ checks should cover every quick question, typed wording variants and a question outside the supported topics. Each result should record what was expected, what appeared and whether a change was needed. This creates a dependable basis for extending the projects.

6.6 Priority 2: User Experience and Accessibility

The next priority is to make each stage of interaction understandable. Processing feedback can tell the user that a request is active; success feedback can confirm the availability of output; and failure guidance can explain how to recover. Labels and controls should work with the keyboard, long content should wrap without hiding essential actions, and text should remain readable at different screen sizes. These improvements should be checked directly rather than inferred from an attractive screenshot. The existing layouts provide a useful foundation for that review.

6.7 Priority 3: Documentation and Extension

Documentation should explain how to open each project, which files perform each role, which translation service is used and which FAQ topics are supported. Future changes can then be assessed against a stable record of behavior. Possible extensions include broader FAQ information, additional translation examples, clearer feedback and better accessibility. These are development directions rather than completed internship features. The overall learning outcome is the ability to connect a user requirement, a structured interface, an observed result and an accurate written explanation.

APPENDIX A
INTERNSHIP COMPLETION CERTIFICATE



Figure A.1 CodeAlpha internship completion certificate

APPENDIX B
INTERNSHIP OFFER LETTER



CODE ALPHA

YOUR DREAM OUR PASSION

INTERNSHIP OFFER LETTER

19th July 2026 **STUDENT ID: CA/DF1/213728**

Name: Sri janani K

Welcome to CodeAlpha

Congratulations**!** We are delighted to make you an offer as **Artificial Intelligence internship** offer letter is to confirm your selection **Artificial Intelligence internship** at **CodeAlpha** and welcome you for the same effective from 20th July 2026 to 20th August 2026. Hope you will be doing well.

This Internship is observed by CodeAlpha as being a learning opportunity for you.

Your internship will embrace orientation and give emphasis on learning new skills with a deeper understanding of concepts through hands-on application of the knowledge you gained as an intern. You will acknowledge your obligation to perform all work allocated to you to the best of your ability within lawful and reasonable direction given to you.

We look forward to a worthwhile and fruitful association which will make you equipped for future projects.

Wishing you the most enjoyable and truly meaningful internship program experience.

Sincerely,

Swati Srivastava
Founder & CEO

CODEALPHA SOFTWARE DEVELOPMENT

GOVERNMENT OF INDIA MSME MINISTRY OF MICRO, SMALL & MEDIUM ENTERPRISES

Phone +91 9336576683

Email services@codealpha.tech

Address Lucknow U.P

Figure B.1 CodeAlpha internship offer letter

APPENDIX C
LETTER OF RECOMMENDATION



CODE ALPHA

Website:
www.codealpha.tech

Email ID:
services@codealpha.tech
support@codealpha.tech

LinkedIn:
www.linkedin.com/company/codealpha

Mobile No:
+91 9336576683

CODEALPHA

YOUR DREAM OUR PASSION

Letter Of Recommendation

TO WHOM IT MAY CONCERN

This is to certify that Sri Janani K has successfully completed the Virtual Internship Program at CodeAlpha for a duration of one month from 20th July 2026 to 20th August 2026 as a Artificial Intelligence Internship.

He/She exhibited performance in this role and made a valuable contribution to our organization during the internship period. He/She has excellent analytical skills and He/She quickly acquired new skills and adeptly adapted to emerging technologies, demonstrating a high level of productivity.

He/She consistently displayed a willingness to provide assistance and fostered strong collaboration with fellow team members. His/Her dedication and skills make him/her a valuable asset to any prospective employer, and CodeAlpha wholeheartedly recommend him/her for any future endeavor.

REGARDS,

CA/DF1/213728

STUDENT I'D

FOUNDER & CEO

CODEALPHA SOFTWARE DEVELOPMENT

20th August 2026

DATE

Figure C.1 CodeAlpha letter of recommendation